UNIT – 5

ASSESSMENT – 3

PROBLEM

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1. SYSTEM DECOMPOSITION & ANALYSIS

Problem

Analyze an existing AI-based system and reverse engineer its architecture by identifying:

- Major system components
- Input and output
- Data flow
- Processing steps
- AI algorithm used
- Interaction between components

Example System: AI Medical Diagnosis System

The system accepts patient symptoms and predicts a possible disease.

2. ARCHITECTURE RECONSTRUCTION

Problem

Given a black-box software system, reconstruct its architecture and explain how different modules interact.

Example

Consider a **black-box AI recommendation system**.

We know:

Input → Recommendation → Output

But the internal architecture is unknown.

3. ALGORITHM IDENTIFICATION

Problem

Reverse engineer a machine learning application to identify:

- Algorithm used
- Data preprocessing
- Features
- Decision-making process
- Prediction logic

Example

Consider a black-box application that predicts whether a student will pass based on study hours and attendance.

4. DATA FLOW & PROCESS MAPPING

Problem

Examine an existing distributed AI application and map:

- Data sources
- Data processing
- AI analysis
- Communication
- Final output

Example

Distributed Fraud Detection System

Transaction Data

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Data Collection

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Data Cleaning

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Feature Extraction

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Distributed Processing

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Fraud Detection

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Alert Generation

5. IMPROVEMENT & REDESIGN

Problem

Perform reverse engineering on an intelligent system and propose improvements to increase:

- Scalability
- Efficiency
- Performance
- Accuracy

Example

Existing system:

User Input

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Single Processing Module

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AI Model

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Result

Problems

1. Single processing unit
2. Slow for large datasets
3. No caching
4. No continuous learning
5. Limited scalability